Example 3.46 Solve $\sin x + \sin 5x = \sin 3x$

Solution:

$\sin x + \sin 5x = \sin 3x \Rightarrow 2\sin 3x \cos 2x = \sin 3x$

$\sin 3x\,(2\cos 2x - 1) = 0$

Thus, either $\sin 3x = 0$ (or) $\cos 2x = \dfrac{1}{2}$

If $\sin 3x = 0$, then $3x = n\pi \Rightarrow x = \dfrac{n\pi}{3}, n \in \mathbb{Z}$ ... (i)

If $\cos 2x = \dfrac{1}{2} \Rightarrow \cos 2x = \cos\dfrac{\pi}{3}$ ... (ii)

$2x = 2n\pi \pm \dfrac{\pi}{3} \Rightarrow x = n\pi \pm \dfrac{\pi}{6}, n \in \mathbb{Z}$

From (i) and (ii), we have the general solution $x = \dfrac{n\pi}{3}$ (or) $x = n\pi \pm \dfrac{\pi}{6}, n \in \mathbb{Z}$.

Example 3.47 Solve $\cos x + \sin x = \cos 2x + \sin 2x$

Solution:

$\cos x + \sin x = \cos 2x + \sin 2x \Rightarrow \cos x - \cos 2x = \sin 2x - \sin x$

$2\sin\left(\dfrac{x+2x}{2}\right)\sin\left(\dfrac{2x-x}{2}\right) = 2\cos\left(\dfrac{2x+x}{2}\right)\sin\left(\dfrac{2x-x}{2}\right)$

$2\sin\left(\dfrac{3x}{2}\right)\sin\left(\dfrac{x}{2}\right) = 2\cos\left(\dfrac{3x}{2}\right)\sin\left(\dfrac{x}{2}\right)$

$\sin\left(\dfrac{x}{2}\right)\left[\sin\left(\dfrac{3x}{2}\right) - \cos\left(\dfrac{3x}{2}\right)\right] = 0$

Thus, either $\sin\left(\dfrac{x}{2}\right) = 0$ (or) $\sin\left(\dfrac{3x}{2}\right) - \cos\left(\dfrac{3x}{2}\right) = 0$.

When $\sin\left(\dfrac{x}{2}\right) = 0$	When $\sin\left(\dfrac{3x}{2}\right) - \cos\left(\dfrac{3x}{2}\right) = 0$
$\dfrac{x}{2} = n\pi \Rightarrow x = 2n\pi, n \in \mathbb{Z}.$	$\Rightarrow \tan\left(\dfrac{3x}{2}\right) = 1 = \tan\left(\dfrac{\pi}{4}\right)$
	$\dfrac{3x}{2} = n\pi + \dfrac{\pi}{4} \Rightarrow x = \dfrac{2n\pi}{3} + \dfrac{\pi}{6}.$

Thus, the general solution is $x = 2n\pi$ (or) $x = \dfrac{2n\pi}{3} + \dfrac{\pi}{6}, n \in \mathbb{Z}$.

If $\sin\theta = \cos\theta$, then $\theta \neq (2n+1)\dfrac{\pi}{2}, n \in \mathbb{Z}$. So, we have $\dfrac{\sin\theta}{\cos\theta} = 1$

Example 3.48 Solve the equation $\sin 9\theta = \sin\theta$

Solution:

$\sin 9\theta = \sin\theta \Rightarrow \sin 9\theta - \sin\theta = 0$

$2\cos 5\theta \sin 4\theta = 0$

Either $\cos 5\theta = 0$ (or) $\sin 4\theta = 0$